

Programme	B. Sc. STATISTICS				
Course Code	STA 8 EJ 416				
Course Title	APPLIED ALGORITHMS AND BIG DATA TECHNIQUES				
Type of Course	Major Elective				
Semester	VIII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Statistical Machine Learning				
Course Summary	To understand how handle big data using EM algorithm, supervisory and unsupervisory learning				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
1.	Explain the concept of EM clustering algorithms.	U	F	Seminar Presentation /
2.	Understand the classification techniques and the concept of support vector machines	U	F	Instructor-created exams
3.	Explain the basic concepts related to big data	Ap	C	Group Tutorial Work
4.	Illustrate the multidimensional scaling techniques in unsupervised learning	Ap	C	Home Assignments
* - Remember (R), Understand (U), Apply (Ap), Analyze (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (60)	Marks (70)
I	EM Algorithm		12	20
	1	Two-Component Mixture Model		
	2	Gaussian Models		
	3	The EM Algorithm in General		
	4	EM as a Maximization–Maximization Procedure		
II	Support Vector Machines		10	15
	5	Maximal Margin Classifier		
	6	Support Vector Classifiers		
	7	Support Vector Machines		

	8	SVMs with More than Two Class- One- Versus-One Classification and One-Versus-All Classification		
III	Big Data		10	15
	9	Definition, Characteristics		
	10	Data Analytics		
	11	General Categories of Data Analytics		
	12	Structured, Unstructured and Semi Structured Data		
	13	Met data		
	14	Big Data Analytics Life Cycle.		
IV	Multi-Dimensional Scaling		16	20
	15	Definition, Perceptual Map		
	16	Decision Frame- work for Perceptual Mapping,		
	17	Non-metric versus Metric methods		
	18	Similarities Data,		
	19	Preferences Data		
	20	Aggregate and Disaggregate Analysis		
	21	De-compositional and Compositional approaches		
	22	Interpreting the MDS results		
V	Open –Ended		12	
	Practical Problems from Module I to Module IV using software's			

Text Books/ References

1. **Hastie, T., Tibshirani, R. and Friedman, J. (2017).** The Elements of Statistical Learning, Data Mining, Inference and Prediction, 2nd edition. Springer, New York.
2. **James, G., Witten, D., Hastie, T. and Tibshirani, R. (2013).** An Introduction to Statistical Learning with Applications in R. Springer, New York.
3. **Erl, T. and Khattak, W. (2016).** Big Data Fundamentals Concepts, Drivers & Techniques. Prentice Hall.
4. **Hair, J. F., Black, W. C., Babin, B. J. and Anderson, R. E. (2009).** Multivariate Data Analysis, 7th edition. Prentice Hall, New York.